

Cyber Security – Unit 4: Tools Used in Cybercrime

Complete 10–15 Marks Exam-Oriented Notes

The topics visible in your syllabus are:

1. **Proxy Servers**
2. **Cybercrime Tools**
3. **Proxy Types**
4. **Proxy-based Cybercrime, Detection & Mitigation**
5. **Anonymizers**
6. **Tor Network Architecture**
7. **Operation Bayonet & AlphaBay Takedown**
8. **Phishing**
9. **Advanced Phishing Techniques**
10. **Password Cracking**
11. **Password Hash Types & Relative Security**
12. **Keyloggers**
13. **Spyware**
14. **Viruses**
15. **Worms**
16. **Virus/Worm Anatomy**
17. **Trojan Horses**
18. **Backdoors**
19. **DDoS Attacks**

I'll explain each in a way you can **directly convert into a 10–15 mark GTU answer**.

1. Proxy Server

Definition

A **proxy server** is an intermediary computer/server that receives requests from a client and forwards those requests to the destination server.

Instead of:

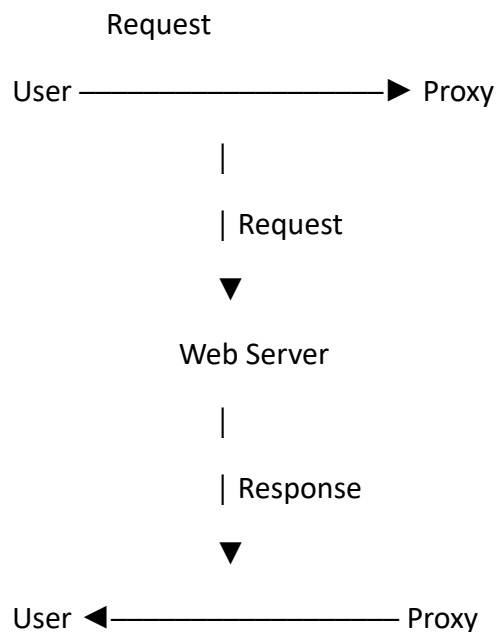
User → Internet Server

the communication becomes:

User → Proxy Server → Internet Server

The destination server generally sees the **proxy's IP address** rather than directly seeing the client's IP address.

Basic Architecture



Functions of Proxy Servers

1. IP masking

A proxy can hide the client's original IP address from the destination server.

2. Access control

Organizations can control which websites or services users can access.

3. Caching

Frequently requested resources can be cached to reduce bandwidth usage.

4. Monitoring

Organizations can monitor network traffic passing through the proxy.

5. Content filtering

Malicious or inappropriate websites can be blocked.

6. Security

A proxy can act as an additional security layer between users and the Internet.

2. Introduction to Cybercrime Tools

Cybercrime tools are software, services, techniques or infrastructure that may be used to perform, facilitate, conceal or automate cyberattacks.

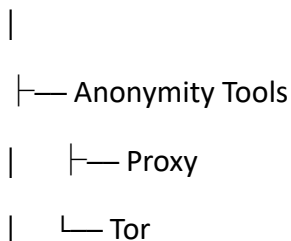
The same technology can sometimes have legitimate security uses.

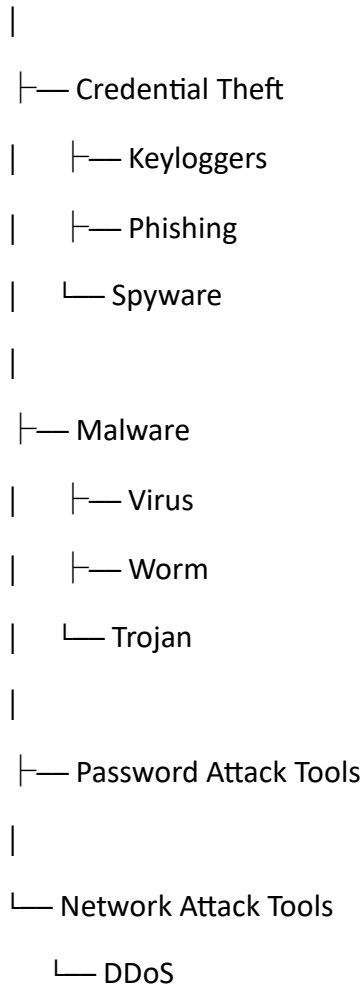
For example:

Tool/Technology	Legitimate Use	Criminal Abuse
Proxy	Privacy/network control	Concealing attack source
Password testing tools	Security auditing	Unauthorized password attacks
Keylogger	Authorized monitoring	Credential theft
Scanner	Vulnerability assessment	Reconnaissance
Encryption	Data protection	Concealing criminal communication
Botnet	Distributed computing research DDoS	

Major categories

Cybercrime Tools





3. Types of Proxy Servers

This is an important exam question.

1. Forward Proxy

A forward proxy represents the **client**.

Client → Forward Proxy → Internet

It is commonly used by organizations to control employee Internet access.

Uses

- Content filtering
- Caching
- Monitoring

- Access control
-

2. Reverse Proxy

A reverse proxy represents the **server**.

Internet → Reverse Proxy → Web Server

It is commonly used by websites and applications.

Uses

- Load balancing
 - Web security
 - SSL/TLS termination
 - Hiding backend infrastructure
 - DDoS protection
-

3. Transparent Proxy

A transparent proxy forwards traffic while generally requiring little or no client-side configuration.

It is commonly used for:

- Network monitoring
 - Content filtering
 - Caching
-

4. Anonymous Proxy

An anonymous proxy attempts to hide the client's IP address from the destination.

5. High-Anonymity Proxy

A high-anonymity proxy provides stronger separation between the client and destination and may avoid revealing identifying proxy information.

6. Open Proxy

An **open proxy** is a proxy accessible by users outside the organization operating it.

Open proxies can be abused by attackers to obscure the source of traffic.

4. Proxy-Based Cybercrime

Attackers may abuse proxy infrastructure to make malicious traffic harder to associate with their original connection.

General concept

Attacker



Proxy 1



Proxy 2



Proxy 3



Target

This is sometimes called **proxy chaining**.

Why criminals use proxies

- Hide source IP
- Make attribution more difficult
- Bypass network restrictions
- Distribute malicious traffic
- Conduct reconnaissance
- Access services from different network locations

Important point

A proxy **does not automatically make someone anonymous**.

Other information may still reveal the source, such as:

- Authentication records
 - Browser/device characteristics
 - Timing information
 - DNS information
 - Application logs
 - Proxy logs
 - Payment records
 - Operational mistakes
-

5. Proxy Detection and Mitigation

Organizations can detect suspicious proxy usage using multiple techniques.

Detection Methods

1. IP reputation

Security systems can identify IP addresses associated with:

- Open proxies
- Known malicious infrastructure
- Botnets
- Abuse networks

2. Traffic analysis

Unusual traffic patterns may indicate proxy or anonymization activity.

3. Authentication monitoring

Repeated logins from geographically unusual locations can be suspicious.

4. DNS analysis

Unexpected DNS behavior can indicate attempts to bypass normal network controls.

5. Log analysis

Security teams can correlate:

Firewall Logs

+

Proxy Logs

+

Authentication Logs

+

Application Logs

Mitigation

- Use firewalls
- Block known malicious IPs
- Apply access-control policies
- Monitor proxy traffic
- Use IDS/IPS
- Apply MFA
- Maintain logs
- Update threat-intelligence lists

6. Anonymizers

Definition

An **anonymizer** is a technology designed to reduce the ability of websites or network observers to directly associate online activity with a user's original network identity.

Examples include:

- Proxy services

- VPNs
- Tor
- Privacy-oriented relay systems

Basic idea

Normal:

User → Website



Website sees user's IP

Anonymized:

User → Anonymizer → Website



Sees intermediary

Types of Anonymizers

1. Proxy-based anonymizer

Uses an intermediary server.

2. VPN

Creates an encrypted connection between the user and VPN infrastructure.

3. Tor

Routes traffic through multiple volunteer-operated relays.

4. Multi-hop systems

Traffic passes through multiple intermediary systems.

7. Technical Functions of Anonymizers

Anonymizers generally provide some combination of:

1. Identity separation

The destination may see the intermediary rather than the original IP.

2. Traffic routing

Traffic is forwarded through intermediary systems.

3. Encryption

Some anonymization systems encrypt traffic between selected network points.

4. Location masking

The destination may see the intermediary's network location rather than the user's.

5. Traffic separation

Multi-hop systems attempt to prevent a single intermediary from learning the complete communication relationship.

8. Tor Network Architecture

Tor = The Onion Router.

Tor is a distributed anonymity network designed to make traffic analysis and direct source identification more difficult.

Why "onion"?

The communication is protected through multiple layers, similar conceptually to layers of an onion.

Basic architecture

TOR NETWORK

User

|



Entry / Guard Relay



Middle Relay



Exit Relay



Internet



Website

Working of Tor

Suppose a user wants to visit a website.

Step 1 – Entry relay

The user's Tor client connects to an entry/guard relay.

Step 2 – Middle relay

Traffic passes through another relay.

Step 3 – Exit relay

Traffic leaves the Tor network through an exit relay.

Step 4 – Destination

The destination website receives the connection from the exit relay.

Onion Routing

Conceptually:

Layer 1 → Entry relay

Layer 2 → Middle relay

Layer 3 → Exit relay

Each relay has limited knowledge of the complete path.

Important components

Component	Function
Tor Client	Runs on user's system
Guard/Entry Relay	First Tor relay
Middle Relay	Intermediate forwarding
Exit Relay	Connects toward normal Internet
Directory System	Helps clients learn relay information

Advantages of Tor

- Privacy
- Resistance to some forms of traffic analysis
- Hides user's IP from destination websites
- Helps users access information privately

Limitations

- Slower than normal browsing
- Exit traffic may be monitored by destination-side observers
- Does not guarantee perfect anonymity
- Endpoint/application leaks can still identify users
- Misconfiguration can expose information

9. Operation Bayonet

Operation Bayonet was a major international law-enforcement operation involving the takedown of major dark-web marketplaces.

Two major marketplaces involved were:

- **AlphaBay**
- **Hansa Market**

The operation demonstrated international cooperation against illegal activities conducted through dark-web marketplaces.

AlphaBay

AlphaBay was a large dark-web marketplace that facilitated illegal transactions.

It was taken down in **2017** as part of international law-enforcement efforts.

Hansa

Hansa was another major dark-web marketplace.

Law enforcement did something particularly significant with Hansa: after taking control of the marketplace, investigators were able to observe activity and gather evidence before the service was ultimately shut down.

Importance of Operation Bayonet

It demonstrated:

1. International law-enforcement cooperation
2. Investigation of dark-web infrastructure
3. Importance of digital evidence
4. Cryptocurrency tracing and financial investigation
5. Operational-security weaknesses

6. Importance of server and infrastructure seizure

Exam conclusion

Operation Bayonet demonstrated that anonymity technologies and dark-web marketplaces do not guarantee immunity from investigation. Traditional investigation, digital forensics, infrastructure analysis and international cooperation can collectively identify and disrupt cybercriminal operations.

10. Phishing

Definition

Phishing is a social-engineering attack in which an attacker attempts to trick a victim into revealing sensitive information or performing an unwanted action by pretending to be a trusted person, organization or service.

Examples of targeted information:

- Username
 - Password
 - OTP
 - Banking information
 - Personal information
 - Authentication tokens
-

11. Phishing Attack Lifecycle

This is a very important **10-mark question**.

Target Selection



Information Gathering



Attack Preparation



Delivery



Victim Interaction



Credential/Data Capture



Unauthorized Use



Covering/Monetization

Step 1 – Target Selection

Attacker chooses individuals or organizations.

Step 2 – Information Gathering

Publicly available information may be used to make fraudulent communication appear convincing.

Step 3 – Preparation

Attacker prepares a deceptive message or fake communication channel.

Step 4 – Delivery

The message may be delivered through:

- Email
- SMS
- Social media
- Messaging platforms
- Fake websites

Step 5 – Victim Interaction

The victim may click a link, open an attachment or respond.

Step 6 – Information Capture

The victim may unknowingly submit sensitive information.

Step 7 – Abuse

Stolen information may be used for:

- Account takeover
- Fraud
- Identity theft
- Further attacks

Step 8 – Concealment

Attackers may attempt to hide or delete evidence.

12. Types of Phishing

1. Email Phishing

Fraudulent emails impersonate legitimate organizations.

2. Spear Phishing

A targeted phishing attack directed toward a specific person or organization.

Example

Instead of sending a generic email to thousands of people, an attacker creates a message specifically designed for an employee.

3. Whaling

Phishing directed toward high-level individuals such as executives.

Employee → Spear Phishing

Executive → Whaling

4. Smishing

Phishing through SMS or text messages.

SMS + Phishing = Smishing

5. Vishing

Phishing through voice calls.

Voice + Phishing = Vishing

6. Clone Phishing

A legitimate-looking message or website is copied and modified to deceive users.

13. Advanced Phishing Techniques

Modern phishing may combine social engineering with technical deception.

1. Social engineering

The attacker manipulates the victim psychologically.

Common principles:

- Urgency
 - Fear
 - Authority
 - Curiosity
 - Reward
 - Trust
-

2. Website impersonation

Attackers create websites that visually resemble legitimate services.

3. Look-alike domains

Domains may be designed to appear similar to legitimate domains.

4. QR-code phishing

Fraudulent QR codes can direct victims toward malicious or deceptive destinations.

5. Multi-stage phishing

A phishing attack may use multiple steps instead of immediately requesting credentials.

Message



Redirect



Fake authentication page



Information collection

Prevention

- Verify sender
- Check domain carefully
- Avoid unexpected links
- Use MFA
- Use password managers
- Verify urgent requests through another communication channel
- Use email security systems
- Train employees
- Report suspicious messages

14. Password Cracking

Definition

Password cracking refers to techniques used to recover or guess passwords, usually by testing possible password values.

In authorized security testing, password auditing is used to evaluate password strength.

15. Password Cracking Methods

1. Brute Force

Attempts possible combinations systematically.

Conceptually:

0000

0001

0002

...

9999

For longer passwords, the number of possibilities becomes extremely large.

Disadvantage

Very time-consuming for strong passwords.

2. Dictionary Attack

Uses a list of likely passwords or words.

For example:

password

admin

welcome

qwerty

...

It is effective against weak, predictable passwords.

3. Hybrid Attack

Combines dictionary words with common modifications.

Conceptually:

password

password1

password123

Password1

4. Credential Stuffing

Attackers use previously leaked username/password combinations against other services.

This works because some users reuse passwords.

5. Password Spraying

Instead of trying many passwords against one account, an attacker tries a small number of commonly used passwords across many accounts.

This can help evade some account-lockout policies.

6. Rainbow Tables

Precomputed data structures can be used against certain unsalted password hashes.

Modern systems can reduce this risk using **salting and strong password hashing**.

16. Popular Password-Cracking/Auditing Tools

For an exam, remember examples:

- **John the Ripper**
- **Hashcat**
- **Hydra**
- **Medusa**
- **Cain & Abel** — historically associated with Windows password recovery/auditing

Important

These tools have legitimate penetration-testing and password-auditing applications, but unauthorized use against accounts or systems is illegal.

17. Password Hashing

A **hash function** converts input into a fixed-length representation.

Password



Hash Function



Hash

A properly designed password-storage system does **not** store passwords directly.

18. Types of Password Hashes and Relative Security

Hash/Method	General Security Context
MD5	Obsolete for password storage
SHA-1	Obsolete for password storage
SHA-256 alone	Not designed specifically for password storage
bcrypt	Designed for password hashing

Hash/Method	General Security Context
scrypt	Memory-hard password hashing
Argon2	Modern password hashing approach
Salted password hash	Stronger against precomputed attacks

Why MD5/SHA-1 are unsuitable for passwords

They are designed to be computationally fast.

For password storage, we actually want hashing to be **deliberately expensive**.

19. Salt

A **salt** is a unique random value added to a password before password hashing.

Password + Unique Salt



Password Hash

Without salt

Two users having the same password may have the same hash.

With unique salts

User A:

Password + Salt A → Hash A

User B:

Password + Salt B → Hash B

Even if both passwords are identical, their stored hashes differ.

20. Keylogger

Definition

A **keylogger** is software or hardware capable of recording keystrokes.

It may be used legitimately in controlled environments, but malicious keyloggers can steal sensitive information.

Information potentially captured

- Usernames
 - Passwords
 - Messages
 - Search terms
 - Financial information
-

21. Types of Keyloggers

1. Software Keylogger

Installed as malicious or unauthorized software.

2. Hardware Keylogger

A physical device placed between a keyboard and computer, or implemented within hardware.

3. Kernel-Level Keylogger

Operates at a low level of the operating system and can be difficult to detect.

4. Browser-Based/Script-Based Logging

Some malicious scripts can capture user input within web applications.

22. Spyware

Definition

Spyware is malicious or unwanted software designed to monitor users and collect information without appropriate consent.

Spyware capabilities

Depending on the type, spyware may:

- Monitor activity
 - Collect browsing information
 - Capture credentials
 - Track applications
 - Collect personal information
 - Record input
 - Communicate information to an external party
-

23. Spyware Distribution

Spyware may be distributed through:

Malicious Attachment



Malicious Software



Drive-by/Deceptive Download



Compromised Application



Social Engineering

Other routes include:

- Fake applications
- Pirated software
- Malicious advertisements
- Compromised websites

- Phishing
-

24. Spyware Detection and Prevention

Detection

- Antivirus/EDR
- Process monitoring
- Network monitoring
- Unexpected startup programs
- Unusual outbound connections
- Security logs

Prevention

- Keep OS updated
 - Use reputable security software
 - Avoid unknown software
 - Use MFA
 - Restrict application privileges
 - Avoid suspicious attachments
 - Regularly review installed applications
-

25. Virus

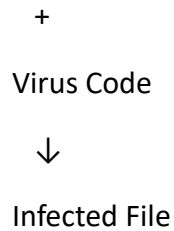
Definition

A **computer virus** is malicious code that can replicate by attaching itself to another file or program and generally requires some form of user/system execution to spread.

Key characteristic

A virus normally needs a host.

Host File



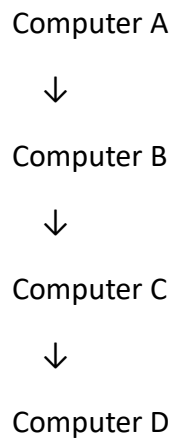
26. Worm

Definition

A **worm** is malware capable of self-replication and spreading between systems, often through networks or vulnerabilities, without requiring a traditional host file.

Key characteristic

A worm is self-propagating.



27. Virus vs Worm

Feature	Virus	Worm
Host required	Usually yes	No traditional host required
Replication	Yes	Yes
Network spreading	May occur	Major characteristic
User interaction	Often required	Often less dependent on user interaction

Feature	Virus	Worm
Main impact	File/system infection	Rapid network/system spread
Speed of spread	Generally slower	Potentially very fast

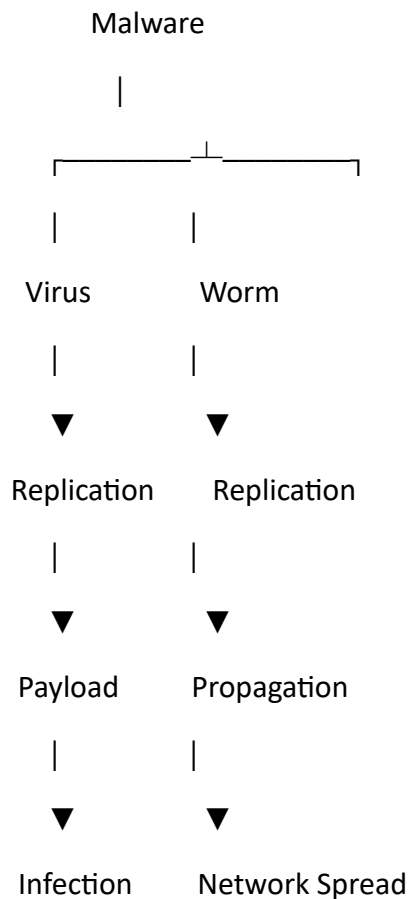
Easy memory trick

Virus = needs a vehicle/host

Worm = walks/spreads by itself

28. Virus/Worm Anatomy

A useful exam diagram:



A malicious program may contain several conceptual components.

1. Replication mechanism

Responsible for creating or propagating copies.

2. Propagation mechanism

Determines how malware moves between files or systems.

3. Payload

The action performed after infection.

Possible consequences include:

- Data destruction
- Data theft
- System disruption
- Installation of additional malware

4. Trigger

A condition that causes a particular action.

Examples:

- Specific date
- System event
- Number of infections

5. Concealment

Some malware attempts to avoid detection.

29. Trojan Horse

Definition

A **Trojan horse** is malware that disguises itself as legitimate or useful software to trick the user into installing or executing it.

Unlike a virus or worm, a Trojan's defining characteristic is **deception**.

Example concept

User believes:

"Useful Application"



Install



Trojan executes

30. Types of Trojans

1. Backdoor Trojan

Creates unauthorized access to a compromised system.

2. Banking Trojan

Targets financial credentials and transactions.

3. Downloader Trojan

Downloads additional malicious software.

4. Spy Trojan

Collects information about the victim.

5. Ransomware Trojan

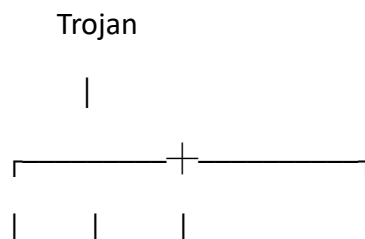
Acts as a delivery mechanism for ransomware.

6. Remote Access Trojan (RAT)

Provides unauthorized remote-control capabilities.

31. Trojan Architecture

Conceptually:



Loader Payload Communication

| | |

| | |

Executes Performs External

Malware Action Server

Components

1. Delivery mechanism

How the Trojan reaches the victim.

2. Loader

Starts or installs the malicious component.

3. Payload

Performs the intended malicious activity.

4. Communication component

May communicate with external infrastructure.

5. Persistence component

May attempt to remain active after system restart.

32. Trojan Detection Methods

1. Antivirus/EDR

Detect suspicious files and behavior.

2. Behavioral analysis

Look for unusual activities such as:

- Unexpected process creation
- Unauthorized persistence
- Suspicious network connections

3. Network monitoring

Detect unusual outbound communication.

4. File integrity monitoring

Detect unauthorized changes to important files.

5. Application allowlisting

Only approved applications are allowed to execute.

33. Backdoors

Definition

A **backdoor** is a mechanism that provides unauthorized or hidden access to a system while bypassing normal authentication or security controls.

Basic concept

Normal Access:

User → Authentication → System

Backdoor:

Attacker → Hidden Access Mechanism → System

34. Types of Backdoors

1. Software Backdoor

Implemented through malicious or vulnerable software.

2. Hardware Backdoor

A hidden mechanism associated with hardware or firmware.

3. Remote Backdoor

Allows unauthorized remote access.

4. Local Backdoor

Provides unauthorized access to users/processes already on the system.

35. Common Backdoor Techniques – High-Level

For your exam, explain conceptually:

1. Hidden accounts

Unauthorized accounts may be created to maintain access.

2. Modified services

A legitimate service may be altered to provide unauthorized functionality.

3. Malicious startup mechanisms

Malware may attempt to execute automatically when a system starts.

4. Remote command channels

A compromised machine may communicate with external infrastructure.

5. Web/application backdoors

Compromised web applications may contain hidden functionality.

36. Backdoor Communication Methods

A backdoor needs some mechanism for receiving commands or sending information.

Conceptually:

Compromised System

|

| Communication



Remote Infrastructure

|

| Commands



Compromised System

Communication may use:

- HTTP/HTTPS
- DNS
- Other network protocols
- Encrypted communication channels

Security systems can identify suspicious patterns through network monitoring.

37. Detection and Prevention of Backdoors

Detection

- Monitor unexpected network connections
- Review user accounts
- Monitor startup programs
- Analyze system logs
- Use EDR
- Check file integrity
- Monitor unusual processes

Prevention

- Patch vulnerabilities
- Use least privilege
- Strong authentication
- MFA
- Network segmentation
- Application allowlisting
- Regular security audits

- Disable unnecessary services

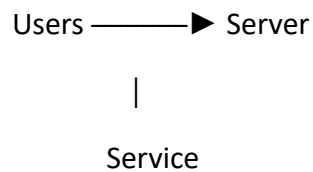
38. DoS Attack

Definition

Denial of Service (DoS) is an attack intended to make a service, server or network resource unavailable or significantly degraded for legitimate users.

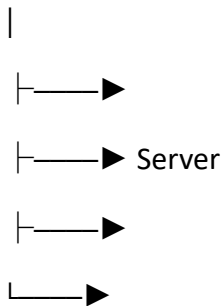
Basic concept

Normal:



During DoS:

Attacker



Server resources exhausted

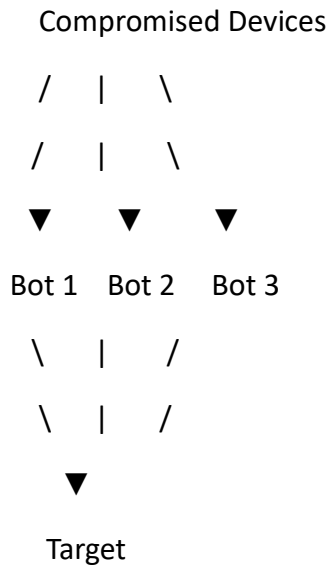


Legitimate users affected

39. DDoS Attack

Definition

Distributed Denial of Service (DDoS) is a DoS attack in which traffic or requests originate from multiple distributed systems.



The distributed nature makes mitigation more difficult.

40. Types of DDoS Attacks

1. Volumetric Attacks

Attempt to consume network bandwidth with large amounts of traffic.

Goal

Available bandwidth



Traffic overload



Service degradation

2. Protocol Attacks

Exploit weaknesses or resource limitations in network/protocol handling.

They may exhaust:

- Connection tables
- Network devices
- Firewall resources

3. Application-Layer Attacks

Target the application itself.

For example, an attacker may generate large numbers of apparently legitimate application requests.

41. DoS vs DDoS

Feature	DoS	DDoS
Full form	Denial of Service	Distributed Denial of Service
Sources	Usually one/fewer sources	Many distributed sources
Coordination	Limited	Often coordinated
Detection	Comparatively simpler	More difficult
Mitigation	Usually simpler	More complex
Traffic origin	Concentrated	Distributed

Easy memory

DoS = One/Few → Target

DDoS = Many → Target

42. DDoS Detection

Organizations can monitor:

Traffic volume

Sudden increases can indicate an attack.

Traffic patterns

Look for abnormal:

- Request rates
- Source distributions
- Protocol behavior
- Geographic patterns

Server performance

Monitor:

- CPU
- Memory
- Connections
- Response time

Network monitoring

Use:

- IDS/IPS
- Firewalls
- Flow monitoring
- SIEM
- DDoS protection services

43. DDoS Mitigation

1. Rate limiting

Restrict excessive requests.

2. Traffic filtering

Block clearly malicious traffic.

3. Load balancing

Distribute traffic among multiple servers.

4. Content Delivery Networks

CDNs can distribute and absorb traffic at large scale.

5. Anycast/network distribution

Traffic can be distributed across multiple network locations.

6. DDoS protection services

Specialized infrastructure can identify and absorb/filter attack traffic.

7. Incident response

Organizations should have a predefined DDoS response plan.

★ Very Important Comparison for Exam

Virus vs Worm vs Trojan vs Backdoor

Feature	Virus	Worm	Trojan	Backdoor
Main characteristic	Infects/replicates	Self-spreads	Deception	Hidden access
Host needed	Usually	No	No	No
Self-replication	Yes	Yes	Usually no	No
Social engineering	Sometimes	Sometimes	Common	Sometimes
Network spreading	Possible	Common	Possible	Possible
Main objective	Infection/damage	Propagation	Gain malicious functionality	Maintain unauthorized access

One-line memory trick

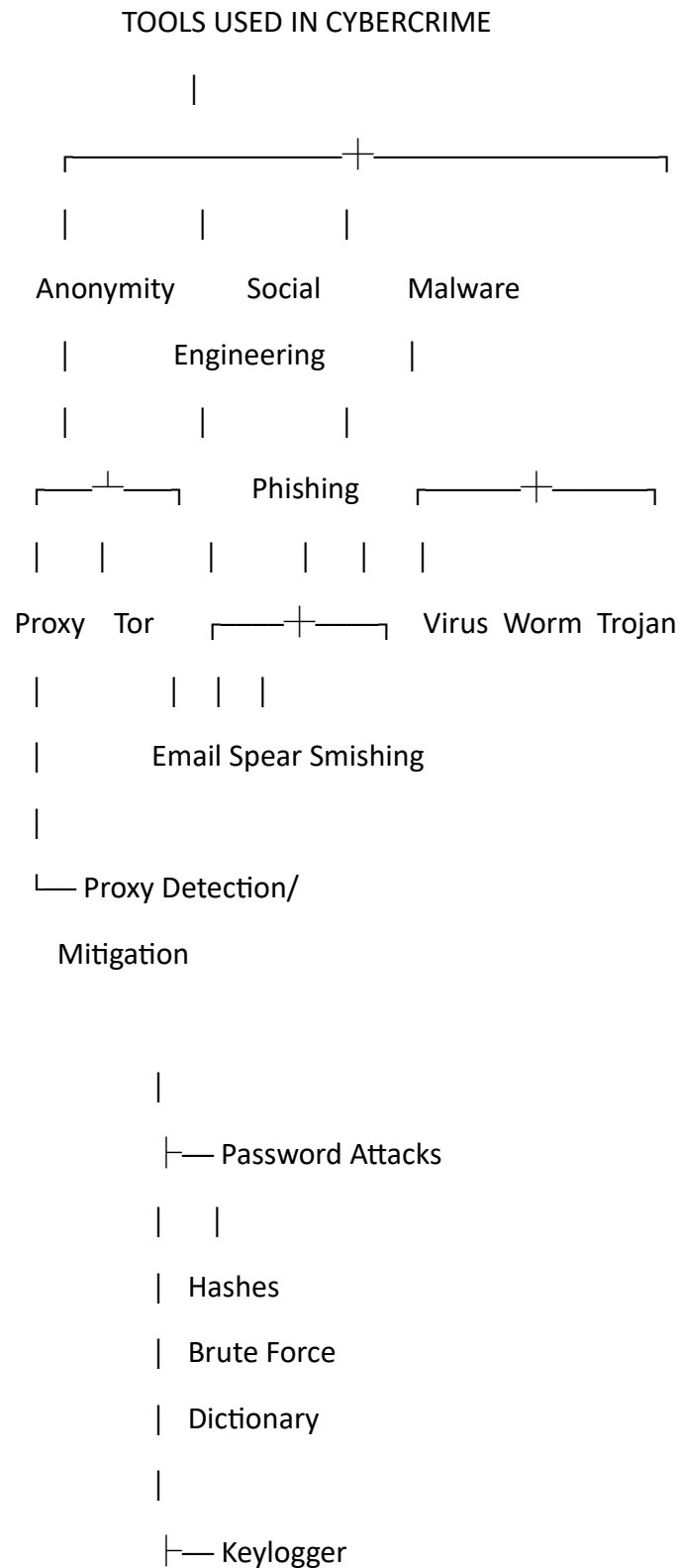
Virus → Infects

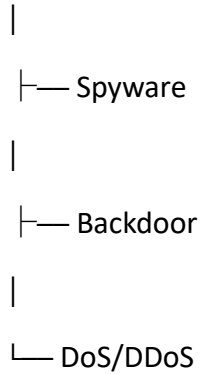
Worm → Spreads

Trojan → Tricks

Backdoor → Enters

★ Complete Unit Mind Map





How to Write a 10–15 Mark Answer in GTU

For almost **any topic from this unit**, use this structure:

1. Definition – 1–2 marks

Give a clear textbook definition.

2. Introduction – 1 mark

Explain why the technology/attack is important.

3. Architecture/Diagram – 2 marks

Always draw a simple diagram.

Example:

Attacker → Intermediate System → Target

4. Working – 3–4 marks

Explain the process step-by-step.

5. Types – 2–3 marks

Give 4–6 types with one-line explanations.

6. Detection/Prevention – 2–3 marks

Especially for cyberattack questions.

7. Advantages/Disadvantages or Impact – 1–2 marks

8. Conclusion – 1 mark

Most Important Questions to Prepare

Based on this portion of your syllabus, make sure you can write full answers for:

1. Explain proxy server, its types and applications.
2. Explain proxy-based cybercrime and its detection/mitigation.
3. Explain anonymizers and their types.
4. Explain Tor network architecture and onion routing.
5. Explain Operation Bayonet and AlphaBay takedown.
6. Explain phishing, its lifecycle and different types.
7. Explain advanced phishing techniques and prevention.
8. Explain password cracking methods.
9. Explain different password hash types and their relative security.
10. Explain keyloggers, their types and prevention.
11. Explain spyware, its capabilities, distribution, detection and prevention.
12. Differentiate virus and worm.
13. Explain virus/worm anatomy.
14. Explain Trojan horse, its types, architecture and detection methods.
15. Explain backdoors, their types and communication methods.
16. Explain DoS and DDoS attacks and their types.
17. Explain DDoS detection and mitigation techniques.
18. Differentiate Virus, Worm, Trojan and Backdoor.

Ultra-short revision trick

Proxy → hides/relays

Tor → multi-hop anonymity

Phishing → tricks user

Password cracking → guesses/recover passwords

Keylogger → records keys

Spyware → secretly monitors

Virus → infects

Worm → spreads

Trojan → disguises

Backdoor → hidden access

DoS → denies service

DDoS → distributed denial of service